UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 7

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Consider $\mathbf{A} \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$.

- Let $\boldsymbol{u}^{(1)}, \boldsymbol{u}^{(2)}, \ldots, \boldsymbol{u}^{(n)}$ be arbitrary eigenvectors, each associated with one eigenvalue.
- Assume $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ are linearly independent, i.e., $\text{span}(\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)})$ has dimension n.
- Then, $\mathbf{A}$ can be decomposed as $\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1}$, where

$$\mathbf{U} = \begin{bmatrix} \boldsymbol{u}^{(1)} & \cdots & \boldsymbol{u}^{(n)} \end{bmatrix} \text{ and } \boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

- Since (by assumption) $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ are linearly independent, $\mathbf{U}$ is invertible.
- In this case, $\mathbf{A}$ is called a *diagonalizable* matrix.
- If $\lambda_1, \ldots, \lambda_n$ are all distinct (not repeated), $\mathbf{A}$ is always diagonalizable.
- If $\mathbf{A}$ has repeated eigenvalues, check Theorem 3.4 in the textbook that explains when such linearly independent eigenvectors $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ exist.

Definition 1 (Orthogonal matrix)**.** A matrix $\mathbf{U} \in \mathbb{R}^{n\times n}$ with columns $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ is called *orthogonal* if

$$\forall i, j, \quad \left\langle \boldsymbol{u}^{(i)}, \boldsymbol{u}^{(j)} \right\rangle = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

This means each column has norm 1 and different columns are orthogonal to each other.

$\mathbf{U}$ is orthogonal iff $\mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I}_n$, where $\mathbf{I}_n$ denotes the identity matrix of dimension $n \times n$.
Thus we have $\mathbf{U} = (\mathbf{U}^{\mathrm{T}})^{-1}$. Also, $\mathbf{U}\mathbf{U}^{\mathrm{T}} = \mathbf{I}_n$.
The inverse of an orthogonal matrix is its transpose: $\mathbf{U}^{-1} = \mathbf{U}^{\mathrm{T}}$.

Example 1. Let $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ be an orthogonal matrix. Then,

$$\mathbf{U}^{-1} = \mathbf{U}^{\mathrm{T}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

■

A matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is called *symmetric* if $\mathbf{A} = \mathbf{A}^{\mathrm{T}}$.

Theorem 1 (Spectral theorem)**.** *Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix, and let $\lambda_1, \ldots, \lambda_n$ be eigenvalues of $\mathbf{A}$. Select an eigenvector $\boldsymbol{u}^{(i)}$ for λ_i $(i = 1, \ldots, n)$ that has length 1. (In the equation $\mathbf{A}\boldsymbol{u} = \lambda\boldsymbol{u}$, you can rescale $\boldsymbol{u}$ to $\alpha\boldsymbol{u}$ to make it length 1.) Then,*

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathrm{T}}.$$

(Here, $\mathbf{U}$ is an orthogonal matrix and $\mathbf{U}^{-1} = \mathbf{U}^{\mathrm{T}}$.)

Therefore, symmetric matrices are always diagonalizable. This plays a big role in optimization through the notion of the Hessian.

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Hessian matrix. Consider $f(\boldsymbol{x})\colon \mathbb{R}^n \to \mathbb{R}$ and assume $f(\boldsymbol{x})$ is twice continuously differentiable:

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}.$$

So, $\nabla^2 f(\boldsymbol{x})$ is an $n \times n$ matrix and the (i,j)-th entry is $\frac{\partial^2 f}{\partial x_i \partial x_j}$. This is called the *Hessian matrix* of f.
Note: due to continuous differentiability,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

Thus, $\nabla^2 f$ is a symmetric matrix.

- If $n = 1$, then $\nabla f(x) = f'(x)$ and $\nabla^2 f(x) = f''(x)$.
- If $f(\boldsymbol{x})$ is quadratic: $f(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x} + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{x} + r$, then

  $$f(\boldsymbol{x}) = \sum_{i=1}^{n} P_{ii}x_i^2 + \sum_{i<j}(P_{ij} + P_{ji})x_i x_j + \sum_{i=1}^{n} q_i x_i + r$$

  and

  $$\frac{\partial f(\boldsymbol{x})}{\partial x_i} = 2\sum_{j=1}^{n} P_{ij}x_j + q_i.$$

  Taking the second-order derivative, we have

  $$\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_i} = 2P_{ii} \quad \text{and} \quad \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} = 2P_{ij}.$$

  Therefore, $\nabla^2 f(\boldsymbol{x}) = 2\mathbf{P}$.

Second-order Taylor series expansion.

- Assume $f(\boldsymbol{x})$ is twice continuously differentiable.
- Given $\boldsymbol{x}_0, \boldsymbol{x} \in \mathbb{R}^n$, we have:

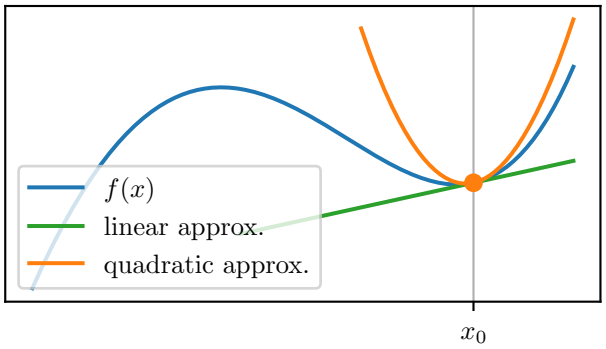
  $$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^{\mathrm{T}}\nabla^2 f(\boldsymbol{x})(\boldsymbol{x} - \boldsymbol{x}_0) + \varepsilon(\boldsymbol{x}),$$

  where the error function $\varepsilon(\boldsymbol{x})$ goes to zero faster than quadratic:

  $$\lim_{\boldsymbol{x} \to \boldsymbol{x}_0} \frac{\varepsilon(\boldsymbol{x})}{\|\boldsymbol{x} - \boldsymbol{x}_0\|^2} = 0.$$

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- $f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0)$ is a linear approximation of $f(\boldsymbol{x})$ at $\boldsymbol{x}_0$.
- But, $f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^{\mathrm{T}}\nabla^2 f(\boldsymbol{x})(\boldsymbol{x} - \boldsymbol{x}_0)$ is a better approximation (quadratic approximation):



- The eigenvalues of $\nabla^2 f(\boldsymbol{x}_0)$ play a big role in studying $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$.

Define $\mathbb{S}^n$ to be the set of all $n \times n$ symmetric matrices. Consider a matrix $\mathbf{A} \in \mathbb{S}^n$.

Definition 2 (Positive semidefinite matrix)**.** A matrix $\mathbf{A} \in \mathbb{S}^n$ is said to be *positive semidefinite* (PSD) if its eigenvalues $\lambda_1(\mathbf{A}), \ldots, \lambda_n(\mathbf{A}) \geq 0$.

Note: the eigenvalues of a symmetric matrix are all real (so they can't be something like $1 + 2\mathrm{i}$ or $-1 + 5\mathrm{i}$), so we are checking if they are positive or not.
Notation: $\mathbf{A}$ being a positive semidefinite matrix is denoted as $\mathbf{A} \succeq \mathbf{0}$.
(We don't show it as $\mathbf{A} \geq 0$ since this means the entries of $\mathbf{A}$ are all non-negative.)

- If $n = 1$, then $\lambda(A) = A$, and the PSD condition $\lambda(A) \geq 0$ is equivalent to $A \geq 0$.
- Thus, PSD is a generalization of non-negativity for scalars.

There is an alternative definition for PSD:

Definition 3 (Positive semidefinite matrix)**.** A matrix $\mathbf{A} \in \mathbb{S}^n$ is said to be *positive semidefinite* (PSD) if

$$\underbrace{\boxed{\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}}}_{\text{scalar}} \geq 0, \quad \forall \boldsymbol{x} \in \mathbb{R}^n.$$

Theorem 2. *Definition 2 and Definition 3 are equivalent.*

Proof. We prove this theorem in two directions.

1. First, we prove that Definition 3 implies Definition 2.

   Assume $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} \geq 0$, $\forall \boldsymbol{x} \in \mathbb{R}^n$.

   Consider an eigenvalue $\lambda_i(\mathbf{A})$ of $\mathbf{A}$, and let $\boldsymbol{u}^{(i)}$ be the corresponding eigenvector. Select $\boldsymbol{x} = \boldsymbol{u}^{(i)}$, then

   $$0 \leq \boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} = (\boldsymbol{u}^{(i)})^{\mathrm{T}}\mathbf{A}\boldsymbol{u}^{(i)} = (\boldsymbol{u}^{(i)})^{\mathrm{T}}\lambda_i(\mathbf{A})\boldsymbol{u}^{(i)} = \lambda_i(\mathbf{A})\|\boldsymbol{u}^{(i)}\|^2.$$

   Since $\|\boldsymbol{u}^{(i)}\|^2 \geq 0$, we have $\lambda_i(\mathbf{A}) \geq 0$ for any i, thus Definition 2 holds.

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2. Then, we prove that Definition 2 implies Definition 3.

   Assume $\lambda_1(\mathbf{A}), \ldots, \lambda_n(\mathbf{A}) \geq 0$. Pick an arbitrary $\boldsymbol{x} \in \mathbb{R}^n$. We need to show that $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} \geq 0$.

   Since $\mathbf{A}$ is symmetric, by the spectral theorem (Theorem 1), there exist an orthogonal matrix $\mathbf{U} \in \mathbb{R}^{n\times n}$ and a diagonal matrix $\boldsymbol{\Lambda} \in \mathbb{R}^{n\times n}$ such that $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathrm{T}}$. Thus

   $$\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} = \boldsymbol{x}^{\mathrm{T}}\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathrm{T}}\boldsymbol{x} = \boldsymbol{y}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{y},$$

   where $\boldsymbol{y}$ is defined as $\mathbf{U}^{\mathrm{T}}\boldsymbol{x}$. It follows that

   $$\begin{aligned}\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} &= \boldsymbol{y}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{y} \\ &= [y_1, \ldots, y_n]\begin{bmatrix}\lambda_1(\mathbf{A}) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n(\mathbf{A})\end{bmatrix}\begin{bmatrix}y_1 \\ \vdots \\ y_n\end{bmatrix} \\ &= \sum_{i=1}^{n} \underbrace{\lambda_i(\mathbf{A})}_{\geq 0}\underbrace{y_i^2}_{\geq 0} \geq 0.\end{aligned}$$

   Therefore, Definition 3 holds.

■

Definition 4 (Positive definite matrix)**.** A matrix $\mathbf{A} \in \mathbb{S}^n$ is said to be *positive definite* (PD) if $\lambda_1(\mathbf{A}), \ldots, \lambda_n(\mathbf{A}) > 0$, or alternatively, if $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} > 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{x} \neq \mathbf{0}$ (note: $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} = 0$ if $\boldsymbol{x} = \mathbf{0}$).

We denote it as $\mathbf{A} \succ \mathbf{0}$.

Definition 5 (Negative semidefinite matrix)**.** A matrix $\mathbf{A} \in \mathbb{S}^n$ is said to be *negative semidefinite* (NSD) if $\lambda_1(\mathbf{A}), \ldots, \lambda_n(\mathbf{A}) \leq 0$, or alternatively, $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} \leq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$.

We denote it as $\mathbf{A} \preceq \mathbf{0}$.

Definition 6 (Negative definite matrix)**.** A matrix $\mathbf{A} \in \mathbb{S}^n$ is said to be *negative definite* (ND) if $\lambda_1(\mathbf{A}), \ldots, \lambda_n(\mathbf{A}) < 0$, or alternatively, if $\boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} < 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{x} \neq \mathbf{0}$.

We denote it as $\mathbf{A} \prec \mathbf{0}$.
A matrix that is not PSD, PD, NSD and ND is called *sign indefinite*, meaning that it has at least one positive and one negative eigenvalue.

Example 2.

- The matrix $\begin{bmatrix}1 & 0 \\ 0 & 2\end{bmatrix}$ is PD and also PSD. (Note that the eigenvalues of a diagonal matrix are its diagonal elements.)
- The matrix $\begin{bmatrix}1 & 0 \\ 0 & 0\end{bmatrix}$ is PSD but not PD.
- The matrix $\begin{bmatrix}-1 & 0 \\ 0 & -2\end{bmatrix}$ is ND and also NSD.

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- The matrix $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$ is NSD but not ND.
- The matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is sign indefinite.

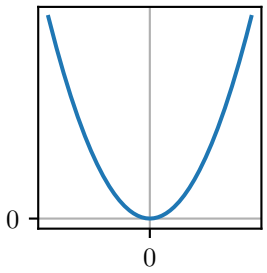
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Recall that if $\boldsymbol{x}^*$ is a local minimum of $f(\boldsymbol{x})$, then $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

- This doesn't imply that if $\nabla f(\boldsymbol{y}) = \mathbf{0}$ for some $\boldsymbol{y}$, then $\boldsymbol{y}$ is a local minimum.
- It is a necessary optimality condition but it is not sufficient.

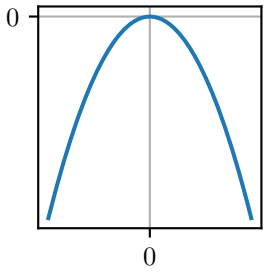
Example 3.

- $\min x^2$.



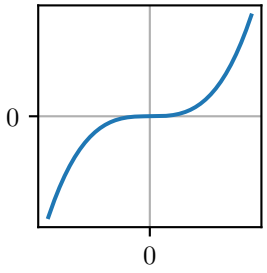
$\nabla f(x) = 0 \Rightarrow x^* = 0$, which is a local minimum.

- $\min -x^2$.



$\nabla f(x) = 0 \Rightarrow x^* = 0$, which is a local maximum, and there is no local minimum.

- $\min x^3$.



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$\nabla f(x) = 0 \Rightarrow x^* = 0$, which is neither minimum nor maximum. It is called a *saddle point.*

■

Theorem 3. *Consider the optimization problem*

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \; f(\boldsymbol{x}).$$

Assume $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$, $\forall \boldsymbol{x} \in \mathbb{R}^n$. *Then,*

(i) All local minima are also global minima.

(ii) $\boldsymbol{x}^*$ *is a local/global minimum iff* $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

Note that $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$ needs to hold for all $\boldsymbol{x}$, not just $\boldsymbol{x}^*$.

Proof. To prove this, recall the Taylor series approximations:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \text{error},$$
$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^{\mathrm{T}}\nabla^2 f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0) + \text{error}.$$

The next theorem removes the error term by adjusting the last term in the right-hand sides:

Lemma 1 (Mid-value theorem)**.** *For each* $\boldsymbol{x} \in \mathbb{R}^n$*, there exist two vectors* $\boldsymbol{u} \in \mathbb{R}^n$ *and* $\boldsymbol{v} \in \mathbb{R}^n$ *on the segment from* $\boldsymbol{x}$ *to* $\boldsymbol{x}_0$ *such that*

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{u})^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0), \tag{1}$$
$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^{\mathrm{T}}\nabla^2 f(\boldsymbol{v})(\boldsymbol{x} - \boldsymbol{x}_0). \tag{2}$$

Illustration of segment when $n = 2$:

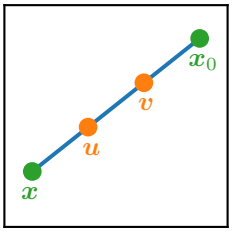
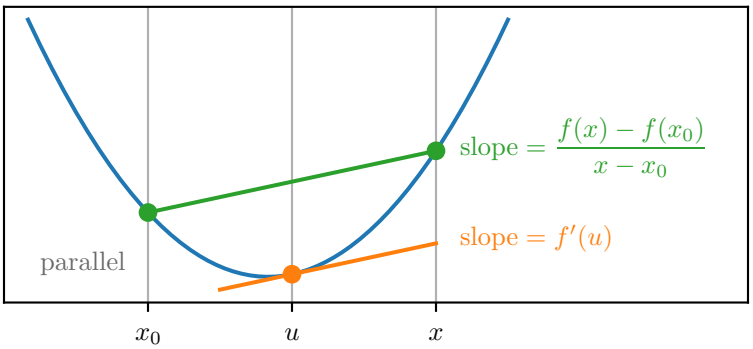


Illustration of Equation (1) for $n = 1$:

$$(1) \Leftrightarrow f(x) = f(x_0) + f'(u)(x - x_0)$$
$$\Leftrightarrow f'(u) = \frac{f(x) - f(x_0)}{x - x_0}.$$

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There is a point u between x and x_0 such that the tangent line of $f(x)$ at u is parallel to the segment between $(x_0, f(x_0))$ and $(x, f(x))$.
Note: the lemma says $\boldsymbol{u}$ and $\boldsymbol{v}$ exist but there is no formula to find them.
Back to the proof of Theorem 3.

- *Necessity proof.* If $\boldsymbol{x}^*$ is a global minimum, it must also be a local minimum. Thus, $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ (this was proven before).
- *Sufficiency proof.* If $\boldsymbol{x}^*$ satisfies $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$, we need to show that $\boldsymbol{x}^*$ is a global minimum (thus also a local minimum).

  Consider an arbitrary point $\boldsymbol{x} \in \mathbb{R}^n$. By Equation (2) of Lemma 1, there exists $\boldsymbol{v} \in \mathbb{R}^n$ such that

$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + \underbrace{\nabla f(\boldsymbol{x}^*)^{\mathrm{T}}(\boldsymbol{x}-\boldsymbol{x}^*)}_{=0 \text{ since } \nabla f(\boldsymbol{x}^*)=\mathbf{0}} + \frac{1}{2}\underbrace{(\boldsymbol{x}-\boldsymbol{x}^*)^{\mathrm{T}}\nabla^2 f(\boldsymbol{v})(\boldsymbol{x}-\boldsymbol{x}^*)}_{\geq 0 \text{ by PSD of } \nabla^2 f}$$

$$\Rightarrow f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*).$$

  Since $\boldsymbol{x}$ is arbitrary, it follows that $\boldsymbol{x}^*$ is a global minimum of f.

■

Let's revisit the least-squares problem:

$$\min \|\mathbf{A}\boldsymbol{x}-\boldsymbol{y}\|^2 = \min \underbrace{\boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x} + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{x} + r}_{f(\boldsymbol{x})},$$

where $\mathbf{P} = \mathbf{A}^{\mathrm{T}}\mathbf{A}$.
Here, $\mathbf{P}$ is PSD because

$$\boldsymbol{x}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A})\boldsymbol{x} = (\mathbf{A}\boldsymbol{x})^{\mathrm{T}}\mathbf{A}\boldsymbol{x} = \|\mathbf{A}\boldsymbol{x}\|^2 \geq 0, \ \forall \boldsymbol{x} \in \mathbb{R}^n.$$

Therefore,

$$\nabla^2 f(\boldsymbol{x}) = 2\mathbf{P} = 2\mathbf{A}^{\mathrm{T}}\mathbf{A} \succeq \mathbf{0}.$$

By Theorem 3, $\boldsymbol{x}^*$ is a global minimum iff $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$. (Recall that $\nabla f(\boldsymbol{x}^*) = 2(\mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^* - \mathbf{A}^{\mathrm{T}}\boldsymbol{y})$.)
To understand the power of previous theorem, let's solve a regularized least-squares problem:

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- Find a point $\boldsymbol{x} \in \mathbb{R}^n$ such that $\mathbf{A}\boldsymbol{x} - \boldsymbol{y}$ is small and the length of $\boldsymbol{x}$ is also small. This is a multi-objective optimization.
- We solve the following optimization problem (called ridge regression):

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2 + \alpha\|\boldsymbol{x}\|^2. \tag{3}$$

  Here, $\alpha \geq 0$ is user-defined and strikes a balance between two objectives:

$$\begin{aligned} \alpha = 0 \quad &\longrightarrow \quad \text{LS}, \\ \alpha \to +\infty \quad &\longrightarrow \quad \boldsymbol{x}^* \to \mathbf{0} \ \ (\|\boldsymbol{x}^*\| \to 0). \end{aligned}$$

  We have

$$\begin{aligned} f(\boldsymbol{x}) &= \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2 + \alpha\|\boldsymbol{x}\|^2 \\ &= \boldsymbol{x}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n)\boldsymbol{x} + (-2\mathbf{A}^{\mathrm{T}}\boldsymbol{y})^{\mathrm{T}}\boldsymbol{x} + (\boldsymbol{y}^{\mathrm{T}}\boldsymbol{y}), \\ \nabla^2 f(\boldsymbol{x}) &= 2(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n). \end{aligned}$$

  Note that

$$\begin{aligned} &\boldsymbol{z}^{\mathrm{T}}\nabla^2 f(\boldsymbol{x})\boldsymbol{z} = 2(\|\mathbf{A}\boldsymbol{z}\|^2 + \alpha\|\boldsymbol{z}\|^2) \geq 0, \ \forall \boldsymbol{z} \in \mathbb{R}^n \\ \Rightarrow\ &\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}. \end{aligned}$$

  By Theorem 3, $\boldsymbol{x}^*$ is a local/global minimum iff $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

  Since

$$\nabla f(\boldsymbol{x}^*) = 2(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n)\boldsymbol{x}^* + (-2\mathbf{A}^{\mathrm{T}}\boldsymbol{y}),$$

  the set of solutions of (3) is

$$\mathcal{S} = \left\{\boldsymbol{x}^* \in \mathbb{R}^n \mid (\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n)\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}\right\}. \tag{4}$$

- Furthermore, if $\alpha > 0$, notice that

$$\begin{aligned} &\boldsymbol{z}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n)\boldsymbol{z} = \underbrace{\|\mathbf{A}\boldsymbol{z}\|^2}_{\geq 0} + \underbrace{\alpha\|\boldsymbol{z}\|^2}_{>0} > 0, \ \forall \boldsymbol{z} \neq \mathbf{0} \\ \Rightarrow\ &\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n \succ \mathbf{0}. \end{aligned}$$

  This means all eigenvalues of $\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n$ are positive, and $\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n$ is invertible. (Note: a matrix is invertible iff it does not have a zero eigenvalue.)

- In this case,

$$\text{Equation (4)} \quad \Rightarrow \quad \boldsymbol{x}^* = (\mathbf{A}^{\mathrm{T}}\mathbf{A} + \alpha\mathbf{I}_n)^{-1}\mathbf{A}^{\mathrm{T}}\boldsymbol{y}$$

  is the unique solution of (3).

To further analyze the least-squares problem, we need the notion of singular values.

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- Recall that a matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ can be decomposed as $\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$ if $\mathbf{A}$ is diagonalizable.
- What if $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $m \neq n$ or $\mathbf{A}$ is not diagonalizable?

Theorem 4 (SVD[1] theorem)**.** *Given an arbitrary matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, there exist matrices $\mathbf{U} \in \mathbb{R}^{m\times m}$, $\mathbf{V} \in \mathbb{R}^{n\times n}$ and $\mathbf{\Sigma} \in \mathbb{R}^{m\times n}$, such that:*

(i) $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$.

(ii) $\mathbf{U}$ *and* $\mathbf{V}$ *are orthogonal matrices (*$\mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I}_m$, $\mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I}_n$*).*

(iii) $\mathbf{\Sigma}$ *is in the form of*

$$\left[\begin{array}{ccc|ccc} \sigma_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_m & 0 & \cdots & 0 \end{array}\right]$$

$$\underbrace{\qquad\qquad}_{m\times m} \quad \underbrace{\qquad\qquad}_{m\times(n-m)}$$

if $n \geq m$, and in the form of

$$\left.\begin{array}{c} \left[\begin{array}{ccc} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n \end{array}\right] \\ \left[\begin{array}{ccc} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{array}\right] \end{array}\right\} \begin{array}{l} n\times n \\ \\ (m-n)\times n \end{array}$$

if $n \leq m$, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

Here, $\sigma_1, \sigma_2, \ldots$ are called the *singular values* of $\mathbf{A}$.

[1] SVD refers to singular value decomposition.

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